Comparison of Finite Difference and Finite Element Methods for Modeling Chloride Diffusion in Concrete

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***Abstract*—A proper understanding of chloride ingress into concrete and effective mitigation of its effects are crucial for preventing corrosion and premature failure of reinforced concrete structures. Previous National Institute of Standards and Technology software employed the finite difference method (FDM) to evaluate chloride diffusion in concrete; however, this approach is not readily extensible to modeling complex geometries including the effects of reinforcing rebar. The purpose of this study is to develop finite element method (FEM) software that can exactly reproduce the FDM results and that can be extended to handle more general and complex geometries, such as concrete containing reinforcing steel. The FEM results show excellent agreement with the FDM results, and the FEM software could therefore be used for more complicated modeling problems.**

***Index Terms*—Chloride, Concrete, Corrosion, Finite Difference Method (FDM), Finite Element Method (FEM)**

I. Introduction

Chloride-induced corrosion of reinforcing steel is the main cause of deterioration of reinforced concrete (RC) structures in chloride-laden environments [1], and, consequently, much research effort has been put towards the prediction of chloride-induced corrosion damage in RC structures [2]–[6]. Specifically, a considerable amount of research has been focused on the chloride-induced corrosion process, and empirical, mathematical, numerical, and probabilistic models have been developed for service life prediction of corrosion-affected RC structures [2], [3], [7]. In the past, both empirical and numerical approaches have been used to develop corrosion rate prediction models, each with its own advantages and limitations. Experimental models work well within the test conditions for which they were created, but they often do not apply effectively to situations beyond the experiments [8]–[10]. Therefore, numerical methods are frequently used to ensure reliable prediction in general conditions. These methods include the finite difference method (FDM) and the finite element method (FEM).

The National Institute of Standards and Technology (NIST) historically deployed FDM software [17], [19] used by researchers to predict chloride diffusion in RC structures [19]. While FDM provides a straightforward and computationally efficient method for modeling diffusion, it is often limited in handling complex geometries and boundary conditions, for which FEM is better suited. Indeed, many papers describe the importance of FEM in chloride diffusion [11], [13], and studies [12] also show that FEM can be used to predict concrete crack initiation and propagation due to chloride-induced corrosion of reinforcing steel. These results are then used to update chloride propagation predictions to realistically represent how concrete cracking and steel corrosion affect each other.

The objective of this research is to develop new FEM software that can exactly reproduce the results of the NIST FDM software and that can be extended to handle more general and complex geometries and boundary conditions. The variational formulation and capacity of FEM to operate on unstructured meshes make it inherently more flexible than FDM. The following sections demonstrate how FEM provides an alternative framework for predicting chloride penetration, which should ultimately inform geometrically precise durability evaluations and maintenance strategies for RC structures. Furthermore, a pathway is identified to translate the existing FDM code developed by NIST into the expanded FEM framework, enabling more accurate and flexible modeling of real-world structural conditions. Presentation of the physics problem is followed by explanations of these numerical methods. The results of the models are then compared and discussed.

II. Physics Problem

Presentation of the physics problem for chloride diffusion involves description of a coupled system of equations and consideration of boundary and initial conditions.

A. Coupled System of Equations

Chloride ingress into concrete is often modeled using the coupled system of equations,

$$\begin{cases} \dfrac{\partial C_{\text{free}}}{\partial t} = \dfrac{\partial}{\partial x}\left(D\,\dfrac{\partial C_{\text{free}}}{\partial x}\right) - k_{\text{react}}\, r\left(C_{\text{react}}\right) C_{\text{free}} \\ \qquad - k_{\text{bind}}\left(\dfrac{\alpha C_{\text{free}}}{1+\beta C_{\text{free}}} - C_{\text{ads}}\right) + f, \\ \qquad 0 \le x \le L,\ 0 < t \le T, \end{cases} \tag{1}$$

$$\frac{dC_{\text{react}}}{dt} = k_{\text{react}} r\left(C_{\text{react}}\right) C_{\text{free}}, \tag{2}$$

$$\frac{dC_{\text{ads}}}{dt} = k_{\text{bind}}\left(\frac{\alpha C_{\text{free}}}{1+\beta C_{\text{free}}} - C_{\text{ads}}\right), \tag{3}$$

where C_{free} denotes the free chloride concentration in the pore solution, C_{ads} describes the adsorbed chloride concentration (mol/m^3), and C_{react} indicates chloride consumed due to irreversible chemical reactions. Known inputs to this system include the diffusion coefficient, D; the reaction rate constant, k_{react}; a reaction indicator function, r, that equals 1 in reactive regions and 0 otherwise; a source term f; and coefficients k_{bind}, α, and β, which control the binding kinetics and adsorption capacity. Here, L denotes the length of the spatial domain, and T indicates the termination time.

Equation (1) represents the diffusion–reaction model governing the free chloride concentration, which is inherently coupled with the amount of free chloride that is consumed due to reactions and adsorption. Equations (2) and (3), consequently, represent irreversible reactions and Langmuir-type adsorption (which describes the binding of chloride ions to the concrete), respectively. By accounting for both the reaction and the adsorption terms, the bound chloride is separated from the total chloride content, and the remaining portion corresponds to the free chloride concentration.

B. Boundary and Initial Conditions

The diffusion–reaction problem is closed by prescribing boundary conditions at both ends of the spatial domain. At the top boundary ($x = 0$), the chloride concentration is typically fixed at a prescribed value, representing a constant or time-dependent exposure condition. This is enforced through a Dirichlet boundary condition,

$$U(0,t) = g, \tag{4}$$

where g denotes the imposed boundary concentration.

At the bottom boundary ($x = L$), chloride transport is frequently controlled through a specified outward flux, which is modeled using a Neumann boundary condition,

$$\left.\frac{\partial U}{\partial x}\right|_{x=L} = -\,h, \tag{5}$$

where h represents the prescribed flux at the boundary, which is typically 0. Together, these boundary conditions reflect a system in which chloride ingress is driven by exposure at one boundary while transport across the opposite boundary is regulated by a specified flux.

To solve the time-dependent diffusion–reaction problem, the initial condition must be specified before starting the time integration. In our case, the initial condition was set to 0 throughout the domain, while a fixed chloride concentration g was prescribed at the top boundary section.

III. Numerical Methods

The coupled diffusion-reaction-adsorption system is solved using a staggered (operator-splitting) time-integration scheme, in which diffusion and reaction/adsorption are solved sequentially at each time step [16]. The Forward Euler method is used for all temporal discretization while we compute the spatial discretization using FDM or FEM. Particularly, we use the Lie splitting [16] in which, for a given time step, the reaction-diffusion system is solved first while neglecting the effects of adsorption, after which the adsorption equation is solved independently using the just-computed results from the reaction-diffusion system. Finally, mass balance is restored by removing the computed adsorbed chloride from the computed free chloride. This splitting technique decouples the nonlinearity at the expense of being first-order accurate in time. However, because Forward Euler is already first-order accurate in time, the effects of this splitting mechanism are consistent with those of our integration scheme, leading to overall efficiency. Both spatial discretization and time discretization are important aspects of FDM and FEM.

A. Finite Difference Method (FDM)

a) Spatial Discretization: The spatial domain is discretized using one-dimensional FDM. The interval $[0, L]$ is divided into N uniform subintervals of size $\Delta x = L/N$, resulting in nodal points $x_i = i\,\Delta x$ for $i = 0, \ldots, N$. The solution at each node is approximated by a discrete variable $d_i(t) \approx d(x_i, t)$.

Spatial derivatives in the diffusion term are approximated using second-order central differences. Boundary conditions are incorporated directly into the discrete system. At the top boundary, the prescribed concentration is enforced by fixing the nodal value, while, at the bottom boundary, the specified flux condition is imposed through a finite difference approximation of the spatial gradient.

b) Time Discretization: Using the Lie splitting, the Forward Euler method for the interior nodes ($1 \le i \le N-1$) gives:

$$d_i^{n+1} = d_i^n + \Delta t\Big[f_i - \frac{D_i(d_i^n - d_{i-1}^n) - D_{i+1}(d_{i+1}^n - d_i^n)}{\Delta x^2} - k_{\text{react}}\, r_i\, d_i^n\Big], \tag{6}$$

With boundary conditions applied explicitly, at the top surface of the concrete slab, the chloride concentration is assumed to be maintained at a fixed value, representing continuous exposure to a chloride-rich environment (considered

as Dirichlet boundary conditions). This condition enforces a constant chloride level at the exposed surface throughout the simulation. At the bottom of the slab, chloride transport is governed by a prescribed flux condition (Neumann boundary conditions) or a constant chloride level (Dirichlet boundary conditions). This represents controlled exchange across the lower boundary, such as limited chloride penetration into an underlying layer. By specifying the rate at which chloride enters or leaves the concrete at this boundary, the model captures the physical constraint on chloride movement at the base of the domain.

Thus, the explicit update at node N is:

$$d_N^{n+1} = d_N^n + \Delta t\Big[f(x_N) + \frac{2h}{\Delta x} + \frac{2D_N}{\Delta x^2}\big(d_{N-1}^n - d_N^n\big) - k_{\text{react}}\, r_N\, d_N^n\Big] \tag{7}$$

c) Courant–Friedrichs–Lewy Condition for Explicit Scheme: To be sure that the explicit method (Forward-Euler) is reliable and the results are stable, the following limitation is considered:

$$\frac{\max(D)\,\Delta t}{\Delta x^2} \leq 0.5 \tag{8}$$

B. Finite Element Method (FEM)

a) Spatial Discretization: The one-dimensional diffusion–reaction problem is solved numerically using FEM with a linear basis function. The formulation follows the standard Galerkin approach described by [14]. FEM is particularly well suited for transport problems in heterogeneous materials such as concrete, as it naturally accommodates spatially varying material properties and allows for local mesh refinement [14].

Making use of the Lie splitting, in abstract form, the weak problem can be written as: find $u \in S$ such that

$$(\dot{u}, w) + (u, k_{react} r w) - a(u, w) = \ell(w) \qquad \forall w \in V, \tag{9}$$

$$a(u, w) = \int_\Omega D\, u_{,x}\; w_{,x}\; dx, \tag{10}$$

$$(u, w) = \int_\Omega u w dx, \tag{11}$$

$$\ell(w) = \int_\Omega f\, w\, dx, \tag{12}$$

where S and V denote appropriate trial and test spaces, $a(\cdot,\cdot)$ is the bilinear form associated with diffusion and reaction processes, $(\dot{u}, w)$ represents the time flux, $(u, k_{react} r w)$ describes the reaction terms, and $\ell(\cdot)$ represents the contribution of source terms and boundary fluxes [14].

The solution field is approximated within a finite-dimensional subspace spanned by shape functions defined over a partition of the spatial domain. This leads to a discrete representation of the form:

$$u_h(x) = \sum_{i=1}^{N} d_i(t)\, N_i(x), \tag{13}$$

where $N_i(x)$ are the finite element shape functions and d_i are the unknown nodal values (coefficients). Substituting this approximation into the weak form and choosing the test functions from the same space yields a system of algebraic equations.

b) Time Discretization: For time-dependent problems, the semi-discrete finite element formulation leads to a system of ordinary differential equations of the form:

$$\mathbf{M}\,\dot{\mathbf{d}}(t) + (\mathbf{K} + \mathbf{R})\,\mathbf{d}(t) = \mathbf{F}(t), \tag{14}$$

where $\mathbf{M}$ is the mass matrix, $\mathbf{K}$ is the diffusion stiffness matrix, $\mathbf{R}$ is the reaction matrix, $\mathbf{d}(t)$ is the vector of time-dependent nodal unknowns, and $\mathbf{F}(t)$ is the load vector.

Using a Forward Euler discretization in time, the time derivative in (14) is approximated as:

$$\dot{\mathbf{d}}(t^n) \approx \frac{\mathbf{d}^{n+1} - \mathbf{d}^n}{\Delta t},$$

where Δt denotes the time-step size and the superscript n indicates the discrete time level.

Substituting this approximation into (14) and solving for $\mathbf{d}^{n+1}$ yields the explicit update formula,

$$\mathbf{d}^{n+1} = \mathbf{d}^n + \Delta t\,\mathbf{M}^{-1}\big(\mathbf{F}^n - (\mathbf{K} + \mathbf{R})\,\mathbf{d}^n\big), \tag{15}$$

which advances the solution from time level t^n to t^{n+1}.

Equation (15) is conditionally stable and requires the time step Δt to satisfy a problem-dependent stability constraint (8).

C. Nitsche Method for Weak Imposition of Dirichlet Boundary Conditions

To weakly impose essential boundary conditions within the finite element framework as is performed in the FDM software, Nitsche's method is employed [15]. This approach allows Dirichlet boundary conditions to be enforced variationally, while maintaining consistency and stability of the numerical scheme. In contrast, FDMs apply boundary conditions directly at grid points, making their implementation simple, especially for structured domains.

In Nitsche's method [15], the standard weak formulation is augmented by additional boundary terms that weakly enforce the Dirichlet boundary condition. These terms ensure consistency with the strong form, restore symmetry of the bilinear form, and add a stabilization contribution controlled by a penalty parameter.

The resulting problem is:

$$(\dot{u}, w) + (u, k_{react} r w) - a_{\text{Nitsche}}(u, w) = \ell_{\text{Nitsche}}(w) \tag{16}$$

where the bilinear form is:

$$a_{\text{Nitsche}}(u, w) = a(u, w) - \int_{\Gamma_D} D\, u_{,x}\; w\, ds - \int_{\Gamma_D} D\, w_{,x}\; u\, ds + \int_{\Gamma_D} \frac{\gamma D}{h_e}\, u\, w\, ds, \tag{17}$$

where u is the trial function, w is the test function, h_e is a characteristic boundary element length, and $\gamma > 0$ is a penalty parameter. A value of $\gamma = 100$ was used, which yielded results

in agreement with the NIST software. The associated linear function is given by:

$$\ell_{\text{Nitsche}}(w) = \ell(w) + \int_{\Gamma_D} D\, w_{,x}\; g\, ds + \int_{\Gamma_D} \frac{\gamma D}{h_e}\, g\, w\, ds, \tag{18}$$

where f is the body force and g denotes the prescribed Dirichlet boundary value.

IV. Comparison of Model Results

In this section, we compare the FEM and FDM formulations for simulating chloride diffusion in concrete. The objectives are threefold: first, to verify that all methods converge; second, to numerically assess the error associated with coarse and fine meshes; and third, to examine the differences between the FEM formulations and the FDM historically used in practice. To make these comparisons, we compute chloride diffusion for a scenario similar to that described for a bridge deck in [17], for which the same domain and parameters are considered for both FDM and FEM. A complete discussion of concrete-chloride chemistry present in these simulations is outside the scope of this paper, but input parameters and forcing behavior for these simulations are described in Tables I and II. The interested reader is referred to [19] for additional guidance.

Because of the numerous types of FEM computations performed, the following notation is followed hereafter for FEM: a plus sign (+) indicates that the corresponding technique is used, while a minus sign (-) indicates that it is not, with potential methods being mass lumping (ML) and Nitsche (N). For instance, FEM(-ML +N) indicates that no mass lumping was used but that Nitsche's method was used.

Comparative plots are first presented for FDM, FEM(-ML -N), and FEM(+ML -N) because each of these strongly enforces input Dirichlet boundary conditions. Coarse mesh computations involved a uniform spatial discretization with 100 spatial elements and 500 time steps for which the results are shown in Fig. 1, while computations on a finer discretization involving 500 spatial elements and 100,000 time steps are shown in Fig. 2. Differences between computed FEM values compared to the FDM benchmark are presented in Figs. 3 and 4 for both the coarse and the fine meshes, respectively. As expected, FEM(+ML -N) results in exactly the same values as the FDM software, while a small difference appears when mass lumping is not used. However, with refinement, this difference diminishes. Specifically, when the differences presented in Figs. 3 and 4 are displayed together, as shown in Fig. 5, a decrease between the FEM and FDM results of approximately one order of magnitude is evident when the fine mesh, rather than the coarse mesh, is used for FEM($\pm$ML -N).

Conversely, results using the Nitsche method differ more dramatically from those with strong boundary enforcement. As shown in Fig. 6, variation in the boundary enforcement parameter γ yields different computed solutions, with higher values of γ more strongly enforcing the top boundary constraint but also introducing worse matrix conditioning and smaller stable time steps. Simulations performed here use 100 uniform spatial elements and 10,000 time steps. The absolute difference between FDM results, each of the Nitsche methods, and the FEM results with strongly enforced boundary conditions are

TABLE I
Model parameters used in the chloride ingress simulations

Symbol	Description
$t_{\max}$	Maximum number of days of exposure: 1800 days
d	Thickness of the concrete: 0.25 m
WC	Water–cement ratio: 0.5
hy	Degree of hydration: 0.75
VF	Volume fraction of aggregate: 70%
AC	Air content (volume fraction): 2.0%
D_i	Time-dependent diffusivity: 6e-12
θ_i	Exterior temperature at month i of exposure: Table II
t_{cur}	Time before exposure begins (s): 28 days
$\kappa_{\text{rel}}^{\text{surf}}$	Relative diffusivity for surface layer: 1
d_{surf}	Thickness of the surface layer: 5.0 mm
E	Activation energy for chloride diffusion: 40 kJ/mole
R	Universal gas constant: 0.008314 kJ/(mol K)
α	Binding isotherm parameter: 1.67
β	Binding isotherm parameter: 4.08
k_{bind}	Binding rate constant: 1e-07
C_3A	C_3A volume fraction (typically 5–10%)
C_4AF	C_4AF volume fraction (typically 5–13%)
C_3A_{react}	Formation of Friedel's salt from C_3A (= 7.419)
C_4AF_{react}	Formation of Friedel's salt from C_4AF (= 4.119)
k_{react}	Chloride–aluminate reaction rate: 1e-08

TABLE II
Monthly chloride concentration and temperature values used in the simulation

Month	Cl^- (mol/m^3)	Temperature (K)
1	4.0000	278.15
2	4.0000	278.15
3	4.0000	288.15
4	4.0000	288.15
5	4.0000	293.15
6	4.0000	298.15
7	4.0000	303.15
8	4.0000	303.15
9	4.0000	298.15
10	4.0000	293.15
11	4.0000	288.15
12	4.0000	283.15

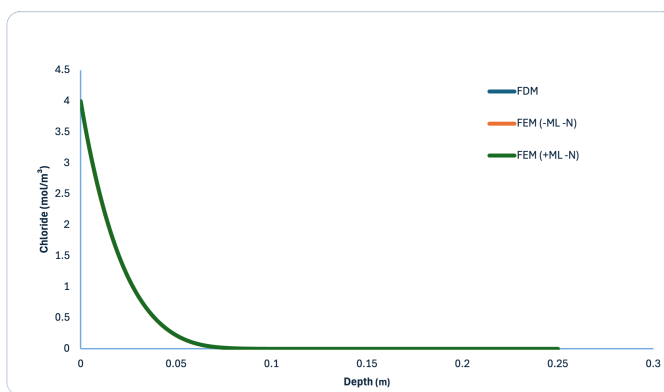

Fig. 1. Comparing FDM and FEM with ML and without Nitsche on the coarse mesh.

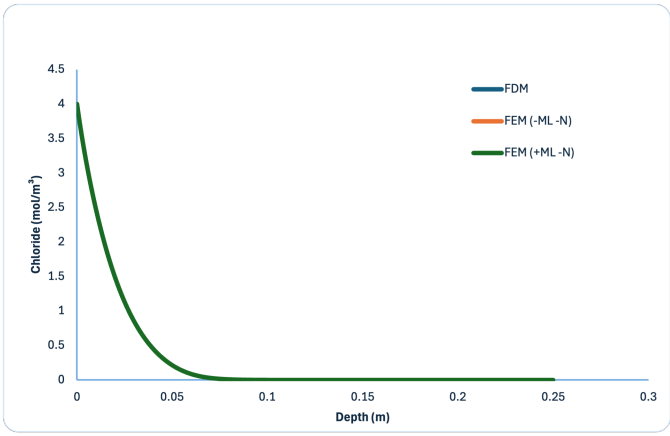

Fig. 2. Comparing FDM and FEM with ML and without Nitsche on the fine mesh.

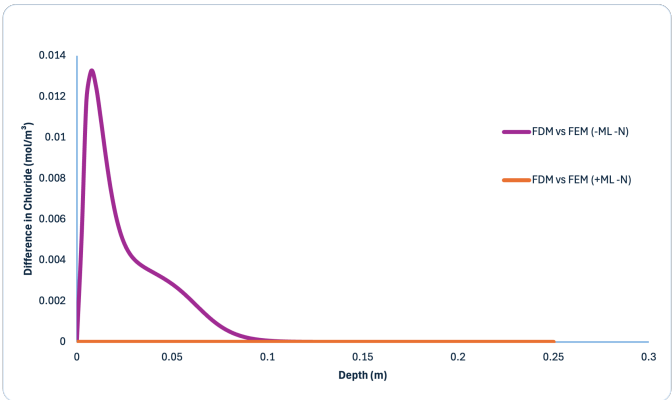

Fig. 3. Error comparison between FDM and FEM formulations on the coarse mesh.

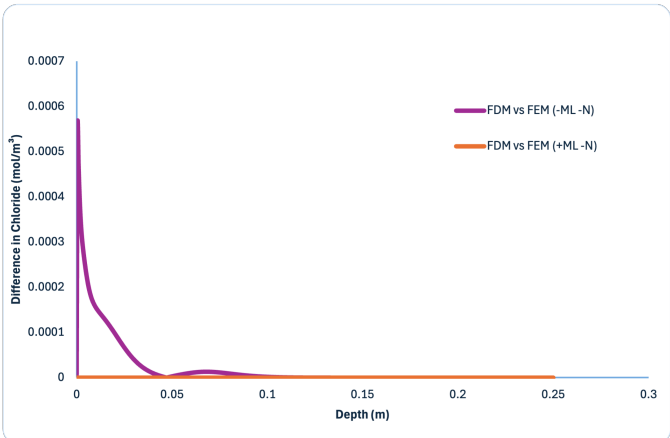

Fig. 4. Error comparison between FDM and FEM formulations on the fine mesh.

shown in Fig. 7. Particularly, Nitsche methods yield results whose difference compared to the FDM results are an order of magnitude larger than that of the FEM results wherein Dirichlet boundary conditions are strongly enforced.

Nitsche methods are also particularly interesting because

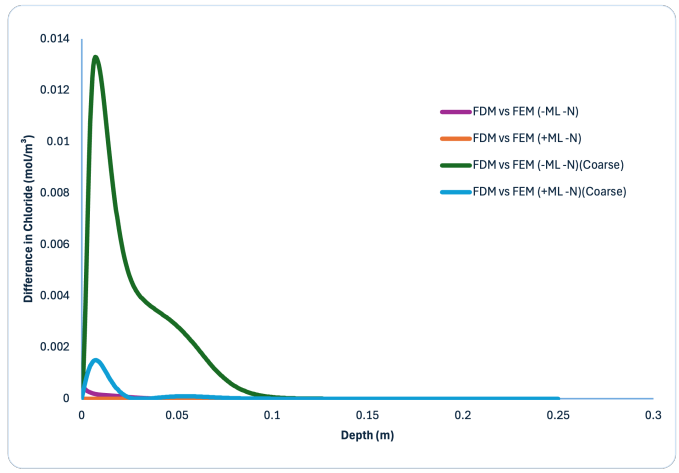

Fig. 5. Error comparison between FDM and FEM formulations on the coarse and fine mesh.

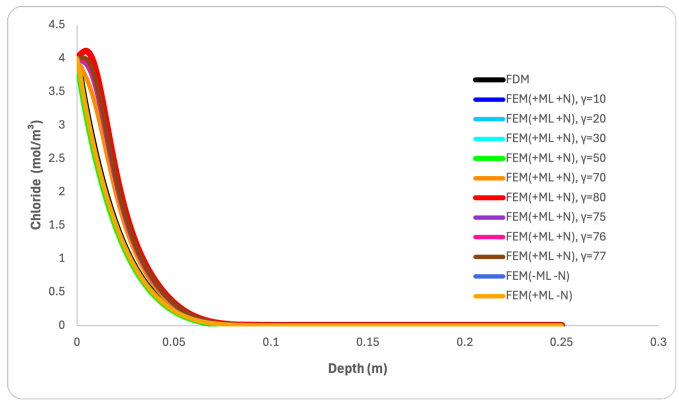

Fig. 6. Comparison between FDM and selected FEM formulations.

stability in the method can still yield non-monotonic solutions, whereas all other solutions are monotonic. As is illustrated in Fig. 6, if $\gamma \geq 80$ is chosen, the maximal chloride value is computed as larger than the input value, making the results less physically meaningful. Consequently, care should be taken when using the Nitsche method to ensure that results are interpreted correctly and match expected concentrations compared to physical measurements.

V. Discussion

Important advantages of FEM over FDM, as well as limitations of both methods are discussed in this section.

A. Advantages of FEM over FDM

FEM offers several advantages over FDM, particularly for engineering applications including complex geometries and boundary conditions. Unlike standard FDM, which is typically restricted to structured grids and regular domains, FEM can naturally accommodate irregular geometries and spatially varying material properties. In addition, FEM provides greater flexibility in the treatment of complex boundary conditions and is generally more suitable for solving problems involving

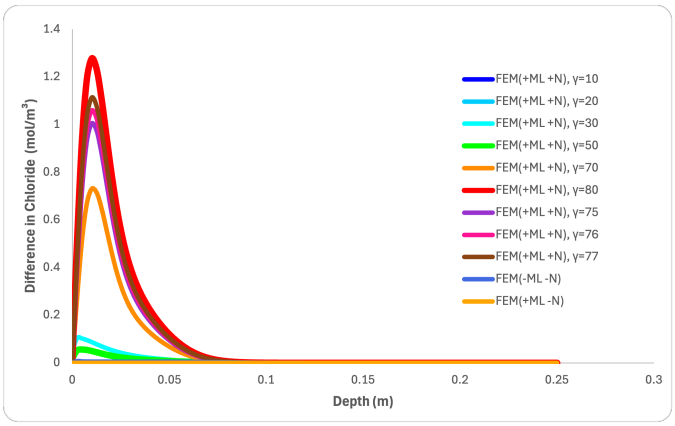

Fig. 7. Error comparison between FDM and selected FEM formulations.

complicated physical processes. As a result, complex geometric problems can often be modeled more effectively using FEM than FDM while still producing the same values in geometrically simple cases. Results also indicate that evaluations using Nitsche's method should be treated with care because a poor choice of the parameter γ may lead to overly weak enforcement of boundary conditions or may lead to non-physical, non-monotonic results.

B. Limitations

Despite their effectiveness, both FEM and FDM require careful treatment of boundary conditions and discretization parameters to ensure numerical accuracy and stability. FDM is computationally efficient and simple to implement, but its applicability is limited for irregular domains and complex geometries. On the other hand, while FEM provides greater flexibility and modeling capability, it typically involves higher computational cost and increased implementation complexity compared to FDM.

VI. Conclusions

In this study, FEM was applied to simulate one-dimensional chloride diffusion, and its performance was compared with FDM. The results show that, with appropriate treatment of boundary conditions, such as the use of Nitsche's method and mass lumping, the FEM solutions closely reproduce the reference FDM results. The results of the FEM(+ML +N) are still comparable for the considered diffusion problem.

In addition to its accuracy, FEM offers greater flexibility for modeling complex engineering problems in RC structures. In particular, it allows the treatment of irregular geometries, spatially varying material properties, and more sophisticated boundary conditions, which are difficult to handle using standard FDM approaches.

Future work could extend FEM analysis for chloride diffusion for 2D and 3D analysis, alternative temporal discretization, or multiphysics extensions to cracking or advective flow.

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